NODE 1

NODE 2

	Type
Red	5V
Orange	Power that isn't GND
Black	GND
Brown	Analog signal
Blue	Parallel bus signal
Green	Serial signals
Yellow	Digital signal
Grey	Control signal

Note that grey signals on the schematic are white IRL

Reset circuitry

+5V, SW3 Reset, R13 10k, R14 330R, RESET_SW, C56 100n, RESET

Crystall oscillator

Y6 4.9152MHz, XTAL1, XTAL2, C55 22p, C58 22p

Power circuitry

U11 LM7805_TO220, +V_IN, -V_IN, C54 1u, C57 10u, +5V

Paralell interface peripherals

U12 ATmega162-16P

PA0	39	PA0
PA1	38	PA1
PA2	37	PA2
PA3	36	PA3
PA4	35	PA4
PA5	34	PA5
PA6	33	PA6
PA7	32	PA7
PB0	1	JOY_BTN
PB1	2	CAN_CS
PB2	3	DISPLAY_D/C
PB3	4	DISPLAY_CS
PB4	5	IO_CS
PB5	6	MOSI
PB6	7	MISO
PB7	8	SCK
PC0	21	PC0
PC1	22	PC1
PC2	23	PC2
PC3	24	PC3
PC4	25	JTAG_TCK
PC5	26	JTAG_TMS
PC6	27	JTAG_TDO
PC7	28	JTAG_TDI
PD0	10	RX
PD1	11	TX
PD2	12	INT
PD3	13	BUSY
PD4	14	CLK
PD5	15	
PD6	16	WR
PD7	17	RD
PE0	31	ALE Lowpass filter
PE1	30	PE1
PE2	29	

RESET, XTAL1, XTAL2, VCC 40, GND 20, ALE, 33R R15, 100n C60, C59 100n

Ekstern minne

PA, PC, CTRL, SRAM, ADC, AIN0–AIN3

IO-Board Connector

JOYSTICK_X	7	8	DISPLAY_RES
JOYSTICK_Y	6	9	PB7
TOUCH_X	5	10	PB6
TOUCH_Y	4	11	PB5
PB0	3	12	PB4
	2	13	PB3
	1	14	PB2

+5V, RESET, IO_Board J18

Serial interface peripherals

Digital Interface

JTAG_TCK, JTAG_TMS, JTAG_TDO, JTAG_TDI, JTAG, CAN, MOSI, MISO, SCK, CS, CANH, CANL, RESET, INT, UART_RX, UART_TX, RS232

Arduino with shield

A1 Arduino Duo

RESET		D7	7
IOREF		D6	6
AREF		D5	5
		D4	4
		D3	3
		D2	2
ADC0		D1/TX	1
ADC1		D0/RX	0
ADC2		D14/TX3	14
ADC3		D15/RX3	15
ADC4		D16/TX2	16
ADC5		D17/RX2	17
ADC6		D18/TX1	18
ADC7		D19/RX1	19
		D20/SDA	20
		D21/SCL	21
		CANRX0	
		CANTX0	

+3V3, VIN, 3V3, +5V, GND

A2 Motor Shield

PHASE/DIR		Motor+	
ENABLE/PWM		+ Motor−	
QUAD_A		Encoder_3V3	
QUAD_B		QUAD_A	
		QUAD_B	
		Encoder_GND	
SERVO_PWM		Servo_PWM	
		Servo_5V	
		Servo_GND	
		CANRX	
		CANTX	

CANH, CANL, +12V, GND

V_IN, V3 1

0 GND, 1 5V, 2 SIGNAL

GND, ChA, ChB, 3V3

Little breadboard

U1C MCP602, C2 100n, C1 100n, R2 1M, U1B MCP602, MCP602 U1A, R5 1k, 330R R4, R3 1k, MPSA56 Q2, R6 100k, R1 47R, V4 1, G5V-2 K1, D4 Snubber

Motorbrett

J3, PWM, Motor_Servo M1, IR Emitter, D1, D2 Photodiode, L1 Solenoide, J1, RotaryEncoder, M2 Motor_DC

NTNU

Gruppe 31

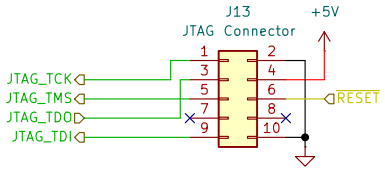
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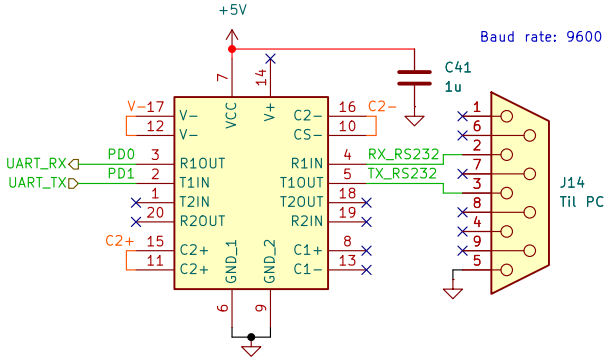
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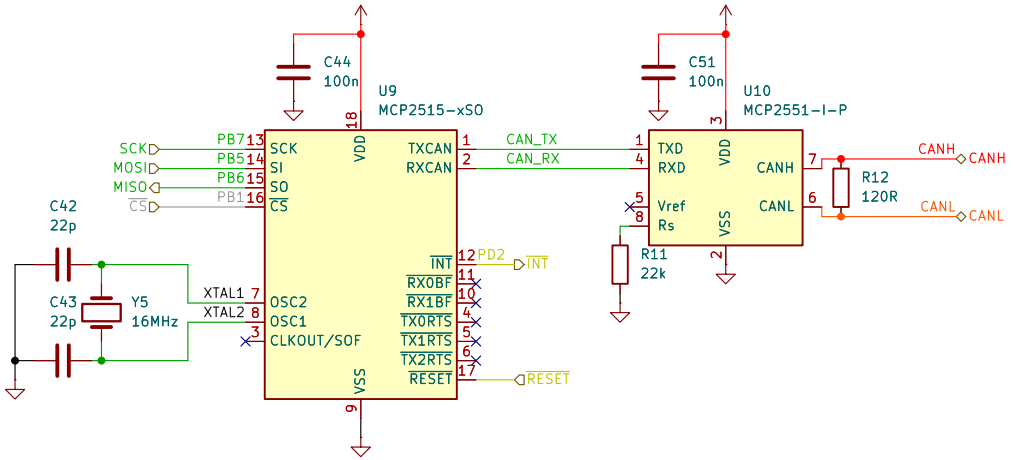
JTAG Interface



RS232 Interface



CAN Interface



Serial interface peripherals

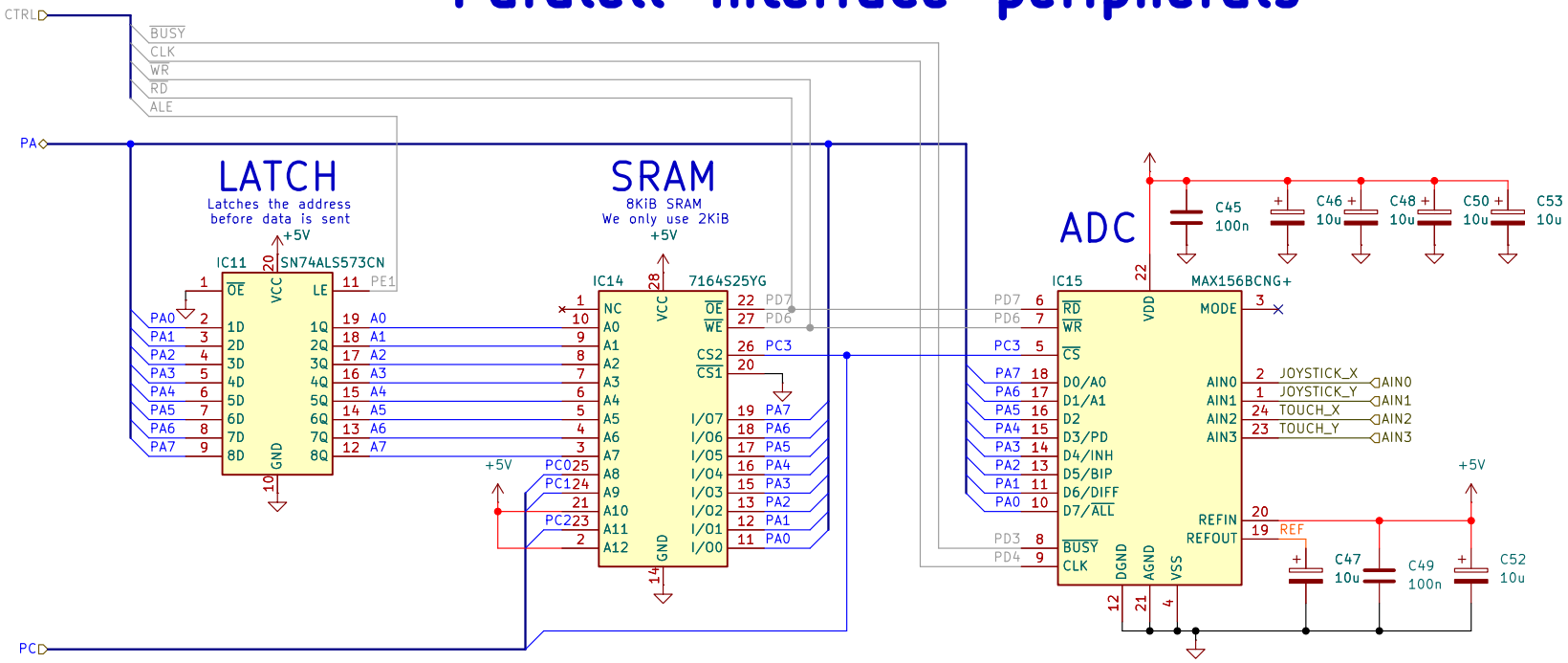
Gruppe 31

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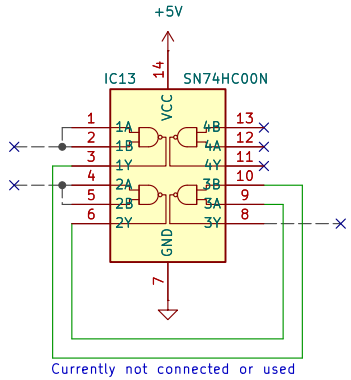
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Paralell interface peripherals



Address decoder



NTNU

Gruppe 31

Sheet: /Ekstern minne/

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Size: A4	Date: 2026-10-07	**Rev: 1 Morbillion**
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